

CARIS DNA + RNA: research findings

TN26-279853 | 5 October 2026, revised overnight | Raw-data research only

MTAP / CDKN2A / CDKN2B. Caris calls all three **Deleted**; CNA values **0.80 / 1.30 / 0.90** are not calibrated tumor copies. MTAP RNA is detectable (4.88 TPM). Expanded analysis does **not** establish homozygous loss or deletion boundaries. **Needed:** vendor CN/LOH segments, purity/ploidy, calibration and an explicit homozygous-loss interpretation.

Coding events and new BAP1 phase evidence. BAP1 c.168_178del, RASA1 c.747delG and APC p.S2577* have mutant RNA support; all 21 selected protein-altering export calls were checked. Near BAP1, joint SNP/deletion pairs give **C+deletion 21**, **C+reference 7**, **G+reference 9**, **G+deletion 0**, independently reproduced. Denser 3p imbalance and local linkage do **not** establish WT-allele loss, somatic status or biallelic inactivation.

BAP1 class-I antigen hypothesis / HLA-B*50:01. Exact mutant windows support **IEERKGLYL** (691 DNA/241 RNA fragments) and overlapping **EERKGLYL** (693/245); MHCflurry binding predictions are 304/364 nM. These are unvalidated antigen candidates. A controlled full-read screen now detects **B*50:01-compatible bulk RNA**, conditional on the earlier RNA-based genotype inference; it proves neither tumor-cell HLA retention nor peptide display. Confirm normal-blood HLA, endogenous presentation and selective recognition using both WT peptides and **31 near-self controls** (seven predict stronger binding than the 8mer).

New RASA1 CD4 testing lead. Natural-terminal **GDYYIGGRRFSSLQT** versus WT **GDYYIGGRRFSSLSD**, provisional DPA1*01:03/DPB1*02:01: CapHLA v2 EL **0.926/0.777**, same ordering in v1. EL is not presentation probability. **Five of seven possible 9mer cores are unchanged WT**; register and selective recognition are unknown. Controls passed; 126 mutant windows were screened with WT controls. No robust mutation-selective BAP1 class-II lead emerged.

Additional LATS candidates. **LATS1 NM_004690.4:c.378dup / p.(Asn127Ter)**: DNA alternate/reference fragments **23/16**, RNA **4/2**; immediate stop after 126 unchanged residues, **no novel tail**. **LATS2 NM_014572.3:c.2878G>A / p.(Gly960Arg)**: DNA **79/38**, RNA **44/5**; function unknown. Separate read/translation audits agree. Both require matched-normal and orthogonal confirmation; Hippo activation or drug sensitivity is not established.

RNA context and expression limits. Re-alignment preserves **246 BAP1 deletion fragments (strict 10-bp flanks)** and mutant-linked annotated splicing; LATS1 remains **4 alternate/2 reference**. Minor BAP1 exon skipping does not establish functional rescue. MSLN **253.01 TPM** needs protein validation. Partial bulk beta CDR3 **CASSLIGGDAFF** has no alpha-beta pairing or tumor-reactivity evidence.

Genome-aware RNA fusion review. All **23.2 million pairs** analyzed against the full reference; **28 accepted + 26 discarded hypotheses** were audited. Many are repeat/normal-RNA signals. CEACAM3 noncoding and KRTAP9-1 repeat events also have DNA support; driver relevance is unproven. CCND1::HEG1 has a small antisense RNA signal and two non-exact DNA read-pair hints; microhomology and overlapping mates limit biological interpretation. **No activating treatment target was established.** Sparse indexing and filtering limit negative conclusions.

Laboratory priorities. Exact GRCh38/RefSeq alleles, WT/mutant FASTAs, read audits and controls are ready. Prioritize calibrated CN/LOH, matched normal, LATS confirmation and controlled presentation/T-cell experiments. Computational reproducibility is distinct from biological or clinical validation.

Source: nine supplied Caris files for TN26-279853; bulk tumor DNA/RNA, no matched normal. Fragment counts collapse paired reads by name, not UMI molecules. Raw-data research only; no pathology, imaging or treatment records used.